B.TECH · 1ST YEAR · 1ST SEMESTER

Engineering Mathematics

Complete Study Module

◈ UNIT III — CALCULUS ◈

Mean Value Theorems & Series Expansions

A complete, step-by-step, exam-oriented guide covering Rolle's Theorem, LMVT, Cauchy's Theorem, Taylor's Theorem, Maclaurin's Theorem, and their Applications — with 40+ fully solved examples and 50 practice problems.

Rolle's Theorem · LMVT · Cauchy's MVT · Taylor's Series · Maclaurin's Series

Approximations · Error Estimation · 40+ Solved Examples · 50 Exercise Problems

Designed for Exam Excellence · Zero Step Skipping · Beginner to Advanced

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1 Introduction to Calculus Theorems

Understanding the Big Picture Before We Dive In

1.1 What is Calculus?

Calculus is the mathematical study of **continuous change**. It has two main branches:

- **Differential Calculus:** Deals with rates of change — how fast something is changing at a given instant.
- **Integral Calculus:** Deals with accumulation of quantities — finding areas, volumes, etc.

In Unit III, we study a special class of theorems called **Mean Value Theorems** — these are powerful results that connect the values of a function to its derivative.

> **Note:** The word "Mean" here means "average." These theorems talk about a point where the derivative equals the average rate of change.

1.2 Why Do We Need These Theorems?

Mean Value Theorems are fundamental tools in mathematics. They help us:

- Prove that a function has a **horizontal tangent** somewhere (Rolle's Theorem)
- Relate the instantaneous rate of change to the **average rate of change** (LMVT)
- Compare two functions and their derivatives (Cauchy's Theorem)
- **Expand functions** as infinite series (Taylor and Maclaurin)
- **Approximate values** like sin(31°), e^0.1, log(1.1), etc.
- Estimate the **error** in approximations

> **Real-world importance:** Engineers use Taylor series to simplify complex equations. Calculators compute sin, cos, log, and e^x using these expansions. Finite element analysis uses these approximations extensively.

1.3 Prerequisites You Must Know

Continuity of a Function

A function f(x) is **continuous** at x = a if:

- f(a) is defined
- lim(x→a) f(x) exists
- lim(x→a) f(x) = f(a)

Simple meaning: You can draw the graph without lifting your pencil.

Differentiability of a Function

A function f(x) is **differentiable** at x = a if:

- f'(a) = lim(h→0) [f(a+h) - f(a)] / h exists
- The left derivative = right derivative

Simple meaning: The graph has no sharp corners or vertical tangents.

⚠️ **Important:** Every differentiable function is continuous, but every continuous function is NOT necessarily differentiable. Example: $f(x) = |x|$ is continuous at $x=0$ but NOT differentiable at $x=0$ (sharp corner).

1.4 The Family of Mean Value Theorems

Theorem	Key Idea	Special Condition
Rolle's Theorem	$f'(c) = 0$ for some c in (a,b)	$f(a) = f(b)$
Lagrange's MVT	$f'(c) = [f(b)-f(a)]/(b-a)$	General case
Cauchy's MVT	$f'(c)/g'(c) = [f(b)-f(a)]/[g(b)-g(a)]$	Two functions
Taylor's Theorem	$f(x)$ as a power series around $x = a$	Infinitely differentiable
Maclaurin's Theorem	Taylor's theorem at $a = 0$	Special case

2

Rolle's Theorem

Statement, Conditions, and Detailed Understanding

2.1 Statement of Rolle's Theorem

ROLLE'S THEOREM

Rolle's Theorem:

Let f(x) be a function defined on a closed interval [a, b]. If the following **three conditions** are satisfied:

Condition 1: f(x) is **continuous** on the closed interval [a, b]
Condition 2: f(x) is **differentiable** on the open interval (a, b)
Condition 3: f(a) = f(b) (the function values at the endpoints are equal)

Then there exists **at least one point c ∈ (a, b)** such that:

```
f '(c) = 0
```

This means the tangent to the curve is horizontal at x = c.

🔵 **In simple words:** If a smooth curve starts and ends at the same height (f(a) = f(b)), then somewhere in between, it must have had a flat moment — a horizontal tangent!

2.2 Understanding Each Condition Deeply

Condition 1: Continuity on [a, b] (Closed Interval)

The function must be continuous at **every point from a to b, including the endpoints** a and b themselves. If the function has a break, jump, or hole anywhere in [a, b], Rolle's theorem does NOT apply.

⚠️ **Common Mistake:** Students write f(x) is continuous on (a, b). But the requirement is on the CLOSED interval [a, b] — endpoints must also be checked!

Condition 2: Differentiability on (a, b) (Open Interval)

The function must be differentiable at **every interior point** — that is, from a to b but NOT including a and b themselves. The endpoints can have sharp corners — only interior points must be smooth.

Condition 3: f(a) = f(b)

The function values at both endpoints must be **exactly equal**. This is the unique condition that separates Rolle's theorem from LMVT.

✅ *Checklist for Rolle's Theorem*

- ✓ f(x) continuous on [a, b]? (Check for polynomials, trig, exponential: usually yes)
- ✓ f(x) differentiable on (a, b)? (No sharp corners inside?)
- ✓ f(a) = f(b)? (Calculate both endpoint values)
- ✓ If ALL THREE ✓, then ∃ c ∈ (a, b) such that f'(c) = 0

How to Find the Value of c?

Once you have verified all three conditions:

1. Find the derivative f'(x)
2. Set f'(x) = 0
3. Solve for x
4. Check that the solution c lies in the open interval (a, b)

2.3 Important Points to Remember

- Rolle's theorem guarantees **existence** — it tells us c exists, but there may be **more than one** such c.
- If the conditions are not met, we say "Rolle's theorem is not applicable" — the theorem doesn't apply, but a point with f'(c) = 0 might still exist accidentally.
- **Polynomials** always satisfy conditions 1 and 2 for any interval.
- **Rational functions** need to be checked for discontinuities (division by zero).
- **Trigonometric functions** like sin x, cos x are always continuous and differentiable everywhere.

3 Verification of Rolle's Theorem

Ultra Step-by-Step Solved Examples

Ex 1 Verify Rolle's Theorem — Basic Polynomial

Verify Rolle's Theorem for $f(x) = x^2 - 5x + 6$ on [2, 3].

STEP 1 – WRITE THE GIVEN FUNCTION AND INTERVAL

Function: $f(x) = x^2 - 5x + 6$

Interval: $[a, b] = [2, 3]$

We need to check all THREE conditions of Rolle's Theorem.

STEP 2 – IDENTIFY WHY ROLLE'S THEOREM MAY APPLY

$f(x) = x^2 - 5x + 6$ is a **polynomial function**.

All polynomial functions are continuous and differentiable **everywhere** on the real line, including any finite closed interval.

So conditions 1 and 2 are automatically satisfied. We only need to verify condition 3.

STEP 3 – VERIFY ALL THREE CONDITIONS

Condition 1 – Continuity on [2, 3]:

$f(x) = x^2 - 5x + 6$ is a polynomial. Polynomials are continuous everywhere.

$\therefore$ $f(x)$ is continuous on [2, 3]. ✓

Condition 2 – Differentiability on (2, 3):

$f(x)$ is a polynomial, so $f'(x)$ exists everywhere.

$\therefore$ $f(x)$ is differentiable on (2, 3). ✓

Condition 3 – Check f(a) = f(b):

$f(2) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = \mathbf{0}$

$f(3) = (3)^2 - 5(3) + 6 = 9 - 15 + 6 = \mathbf{0}$

Since $f(2) = 0 = f(3)$, condition 3 is satisfied. ✓

All three conditions are met. Therefore, Rolle's Theorem is applicable.

STEP 4 – WRITE THE CONCLUSION OF ROLLE'S THEOREM

By Rolle's Theorem, there exists at least one $c \in (2, 3)$ such that:

```
f '(c) = 0
```

STEP 5 – FIND F '(X) BY DIFFERENTIATING STEP-BY-STEP

$f(x) = x^2 - 5x + 6$

Differentiating term by term:

- $d/dx\,(x^2) = 2x$ (using power rule: $d/dx(x^n) = n \cdot x^{n-1}$)
- $d/dx\,(-5x) = -5$ (derivative of constant × x is just that constant)
- $d/dx\,(6) = 0$ (derivative of any constant is zero)

$\therefore f'(x) = 2x - 5$

STEP 6 – SET F '(C) = 0 AND SOLVE FOR C

We set $f'(c) = 0$:

$2c - 5 = 0$

$2c = 5$

$c = 5/2 =$ **2.5**

STEP 7 – VERIFY THAT C LIES IN (A, B)

We found $c = 2.5$.

Check: Is $2.5 \in (2, 3)$?

Since $2 < 2.5 < 3$, YES — $c = 2.5$ is indeed in the open interval $(2, 3)$. ✓

STEP 8 – FINAL RESULT

$c = 5/2 = 2.5$ is the point at which $f'(c) = 0$.

STEP 9 – GEOMETRICAL INTERPRETATION

The parabola $f(x) = x^2 - 5x + 6$ touches both $x = 2$ and $x = 3$ at height $y = 0$ (the x-axis).

At $x = 2.5$ (the vertex of the parabola), the tangent line is perfectly horizontal (slope = 0).

This is exactly what Rolle's theorem predicted!

✅ **Rolle's Theorem is verified. $c = 5/2 = 2.5 \in (2, 3)$ and $f'(2.5) = 0$**

💡 **Key Takeaway:** For polynomials, conditions 1 and 2 are always satisfied. The key step is checking $f(a) = f(b)$. Then differentiate, set equal to zero, solve, and verify c is inside the interval.

Ex 2 Rolle's Theorem — Cubic Polynomial

Verify Rolle's Theorem for $f(x) = x^3 - 6x^2 + 11x - 6$ on $[1, 3]$.

STEP 1 – GIVEN

$f(x) = x^3 - 6x^2 + 11x - 6$, Interval: $[1, 3]$

STEP 3 – VERIFY CONDITIONS

Condition 1: Polynomial → Continuous on [1, 3]. ✓

Condition 2: Polynomial → Differentiable on (1, 3). ✓

Condition 3:

$f(1) = (1)^3 - 6(1)^2 + 11(1) - 6 = 1 - 6 + 11 - 6 = \mathbf{0}$

$f(3) = (3)^3 - 6(3)^2 + 11(3) - 6 = 27 - 54 + 33 - 6 = \mathbf{0}$

$f(1) = f(3) = 0$. ✓ All conditions satisfied.

STEP 5 – FIND F'(X)

$f(x) = x^3 - 6x^2 + 11x - 6$

- $d/dx(x^3) = 3x^2$ (power rule)
- $d/dx(-6x^2) = -12x$
- $d/dx(11x) = 11$
- $d/dx(-6) = 0$

$\therefore f'(x) = 3x^2 - 12x + 11$

STEP 6 – SOLVE F'(C) = 0

$3c^2 - 12c + 11 = 0$

Using the quadratic formula: $c = [12 \pm \sqrt{(144 - 132)}] / 6 = [12 \pm \sqrt{12}] / 6$

$\sqrt{12} = 2\sqrt{3} \approx 3.464$

$c = (12 + 3.464)/6 \approx 2.577$ or $c = (12 - 3.464)/6 \approx 1.423$

Both $c_1 \approx 1.423$ and $c_2 \approx 2.577$ lie in $(1, 3)$. ✓

✅ **Rolle's Theorem verified. $c = (6 \pm \sqrt{3})/3$ — Two values of c exist in (1, 3).**

💡 **Key Takeaway:** Rolle's theorem says "at least one c exists" — there may be MORE than one. Here we found two such points!

Ex 3 Rolle's Theorem — Trigonometric Function

Verify Rolle's Theorem for $f(x) = \sin x$ on $[0, \pi]$.

STEP 1

$f(x) = \sin x$, $[a, b] = [0, \pi]$

STEP 3 – VERIFY CONDITIONS

Condition 1: $\sin x$ is continuous everywhere, so it is continuous on $[0, \pi]$. ✓

Condition 2: $\sin x$ is differentiable everywhere, so differentiable on $(0, \pi)$. ✓

Condition 3:

$f(0) = \sin(0) = \mathbf{0}$

$f(\pi) = \sin(\pi) = \mathbf{0}$

$f(0) = f(\pi) = 0$. ✓

STEP 5 – DIFFERENTIATE F(X)

$f(x) = \sin x$

$f'(x) = \cos x$

(Standard result: derivative of $\sin x$ is $\cos x$)

STEP 6 – SOLVE F'(C) = 0

$\cos c = 0$

$c = \pi/2$ (in the interval $(0, \pi)$)

Since $0 < \pi/2 < \pi$, $c = \pi/2 \in (0, \pi)$. ✓

STEP 9 – GEOMETRIC MEANING

The sine curve from 0 to π starts at 0, rises to 1 at $x = \pi/2$ (the peak), then falls back to 0.

At the peak ($x = \pi/2$), the tangent is horizontal. This is where $f'(c) = 0$.

✅ **$c = \pi/2 \in (0, \pi)$, $f'(\pi/2) = \cos(\pi/2) = 0$**

4 Geometrical Interpretation of Rolle's Theorem

Seeing the Theorem Through a Graph

4.1 The Geometric Picture

Consider a curve $y = f(x)$ on the interval $[a, b]$. If the curve:

- Has no breaks (continuous)
- Has no sharp corners or kinks (differentiable)
- Starts and ends at the SAME HEIGHT ($f(a) = f(b)$)

Then somewhere in between, the curve must have a **horizontal tangent** (the tangent line is parallel to the x-axis).

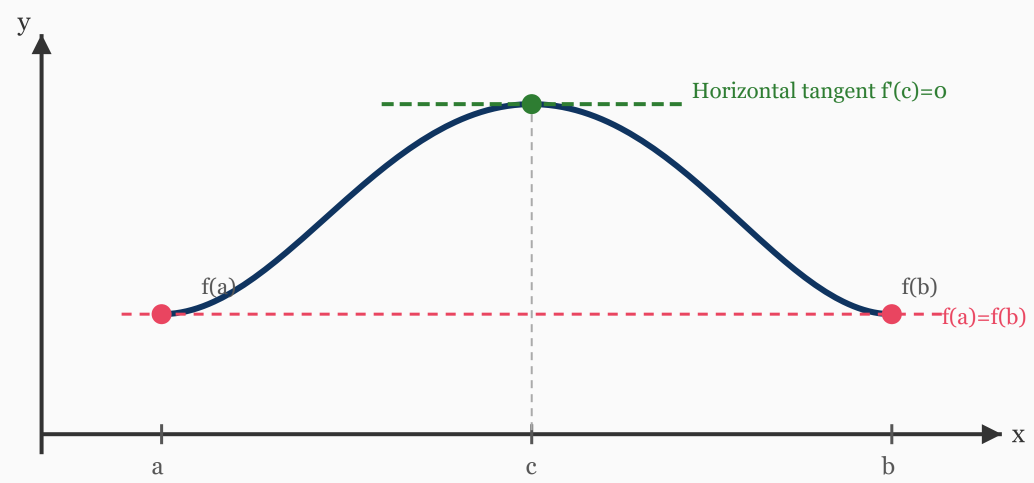

Fig 4.1: Rolle's Theorem — The curve starts and ends at the same height. At $x = c$, the tangent is horizontal ($f'(c) = 0$)

🔵 Why must there be a horizontal tangent?

Think of it physically: Imagine a ball thrown in the air. If it starts and ends at the same height, it must have reached a highest point (or lowest point) somewhere in between. At that turning point, the velocity (derivative) is zero. That's exactly what Rolle's theorem says!

4.2 What Happens When Conditions Fail?

⚠️ Case A — Discontinuity:

If f has a jump at some point, the curve is broken. There may be no horizontal tangent in the interval.

⚠️ Case B — Not Differentiable:

If f has a sharp corner (like $|x|$), the curve may not have a smooth horizontal tangent even if $f(a) = f(b)$.

5

Lagrange's Mean Value Theorem (LMVT)

The Generalization of Rolle's Theorem

5.1 Statement of LMVT

LMVT

Lagrange's Mean Value Theorem:

Let f(x) be a function defined on [a, b] such that:

Condition 1: f(x) is **continuous** on [a, b]

Condition 2: f(x) is **differentiable** on (a, b)

Then there exists at least one point c ∈ (a, b) such that:

```
f '(c) = [f(b) − f(a)] / (b − a)
```

The derivative at c equals the average rate of change of f over [a, b].

Relationship with Rolle's Theorem:

LMVT is a GENERALIZATION of Rolle's Theorem. If f(a) = f(b), then:

f'(c) = [f(b) − f(a)] / (b − a) = 0 / (b − a) = 0

This is exactly Rolle's theorem! So Rolle's theorem is a SPECIAL CASE of LMVT.

5.2 Understanding the Formula

The right side of the formula **[f(b) − f(a)] / (b − a)** is the **slope of the chord** joining the points (a, f(a)) and (b, f(b)).

The left side **f'(c)** is the **slope of the tangent** at x = c.

LMVT says: There is a point c where the tangent line is *parallel* to the chord joining the endpoints!

5.3 Conditions Comparison: Rolle's vs LMVT

Condition	Rolle's Theorem	LMVT
Continuity on [a,b]	✓ Required	✓ Required
Differentiability on (a,b)	✓ Required	✓ Required
f(a) = f(b)	✓ Required	✗ NOT Required
Conclusion	f'(c) = 0	f'(c) = [f(b)−f(a)]/(b−a)

Ex 4 LMVT — Basic Polynomial

Verify LMVT for $f(x) = x^2$ on [1, 3].

STEP 1 – GIVEN

$f(x) = x^2$, $[a,b] = [1,3]$

STEP 3 – VERIFY CONDITIONS

Condition 1: $f(x) = x^2$ is a polynomial → continuous on [1,3]. ✓

Condition 2: $f(x) = x^2$ is a polynomial → differentiable on (1,3). ✓

(Note: For LMVT we do NOT need to check $f(a) = f(b)$!)

$f(1) = 1$, $f(3) = 9$ — they are NOT equal, but LMVT still applies!

STEP 4 – WRITE THE LMVT FORMULA

There exists $c \in (1,3)$ such that:

$f'(c) = [f(3) - f(1)] / (3 - 1) = [9 - 1] / 2 = 8/2 = \mathbf{4}$

STEP 5 – FIND F'(X)

$f(x) = x^2$

$d/dx(x^2) = 2x$ (power rule: bring the power down, reduce power by 1)

$\therefore f'(x) = 2x$

STEP 6 – SET F'(C) = 4 AND SOLVE

$2c = 4$

$c = 2$

Check: Is $c = 2 \in (1, 3)$? Yes, since $1 < 2 < 3$. ✓

✅ **$c = 2 \in (1,3)$, $f'(2) = 4 = [f(3)-f(1)]/(3-1)$**

💡 **Meaning:** At $x = 2$, the slope of the tangent to $y = x^2$ equals the slope of the chord from (1,1) to (3,9). Both are 4.

Ex 5 LMVT — Logarithmic Function

Verify LMVT for $f(x) = \log x$ on $[1, e]$.

STEP 1

$f(x) = \log x$ (natural logarithm), $[a,b] = [1, e]$

STEP 3 – VERIFY CONDITIONS

Condition 1: $\log x$ is continuous for $x > 0$. Since $[1, e] \subset (0, \infty)$, $\log x$ is continuous on $[1, e]$. ✓

Condition 2: $f'(x) = 1/x$ exists for $x > 0$. Since $(1, e) \subset (0, \infty)$, f is differentiable on $(1, e)$. ✓

STEP 4 – APPLY LMVT FORMULA

$f(1) = \log(1) = 0$

$f(e) = \log(e) = 1$

Average rate of change $= [f(e) - f(1)] / (e - 1) = [1 - 0] / (e - 1) = 1/(e - 1)$

STEP 5 – FIND F'(X)

$f(x) = \log x$

$f'(x) = 1/x$ (standard derivative of natural log)

STEP 6 – SET F'(C) = 1/(E–1)

$1/c = 1/(e - 1)$

$c = e - 1 \approx 1.718$

Check: Is $c = e-1 \in (1, e)$? Since $1 < 1.718 < 2.718 = e$, YES! ✓

✅ **$c = e - 1 \approx 1.718 \in (1, e)$**

Ex 6 LMVT — Cubic with detailed algebra

Find c for $f(x) = x^3 - x$ on $[0, 2]$ using LMVT.

STEP 3 – CONDITIONS

Polynomial → Continuous on $[0,2]$ ✓ and Differentiable on $(0,2)$ ✓

STEP 4 – LMVT CALCULATION

$f(0) = 0^3 - 0 = 0$

$f(2) = 2^3 - 2 = 8 - 2 = 6$

$[f(2) - f(0)] / (2 - 0) = 6 / 2 = 3$

STEP 5 – DIFFERENTIATE F(X)

$f(x) = x^3 - x$

$f'(x) = 3x^2 - 1$

STEP 6 – SOLVE

$3c^2 - 1 = 3$

$3c^2 = 4$

$c^2 = 4/3$

$c = \pm 2/\sqrt{3}$

$c = 2/\sqrt{3} \approx 1.155$ (taking positive value since interval is $[0,2]$)

Check: $0 < 1.155 < 2$. ✓

✅ **$c = 2/\sqrt{3} = 2\sqrt{3}/3 \approx 1.155 \in (0, 2)$**

6

Geometrical Interpretation of LMVT

Seeing the Theorem as Parallel Lines

6.1 The Geometric Picture of LMVT

The chord joining (a, f(a)) and (b, f(b)) has slope = [f(b) – f(a)] / (b – a).

LMVT says: There is a point c where the tangent to the curve has the **same slope as this chord** — meaning the tangent is **parallel to the chord**.

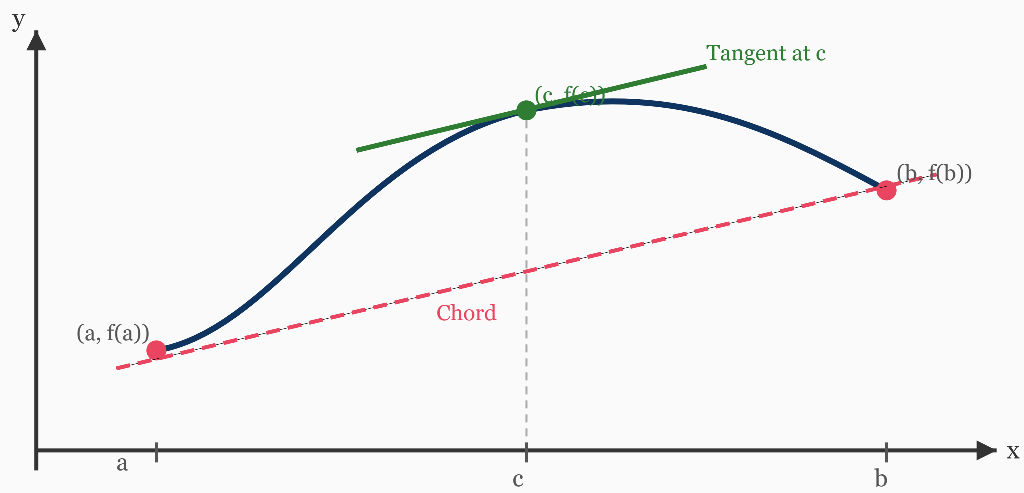

Fig 6.1: LMVT — At x = c, the tangent (green) is parallel to the chord (red dashed) joining the endpoints. Both have the same slope.

🔵 **Physical Meaning:** If you travel from city A to city B, your average speed is the total distance divided by total time. LMVT says: at some point during the journey, your *instantaneous speed* was EXACTLY EQUAL to the average speed!

7 Cauchy's Mean Value Theorem

LMVT for Two Functions Simultaneously

7.1 Statement of Cauchy's Mean Value Theorem

CAUCHY

Cauchy's Mean Value Theorem:

Let f(x) and g(x) be two functions defined on [a, b] such that:

Condition 1: Both f(x) and g(x) are **continuous** on [a, b]

Condition 2: Both f(x) and g(x) are **differentiable** on (a, b)

Condition 3: $g'(x) \neq 0$ for any x in (a, b)

Then there exists at least one $c \in (a, b)$ such that:

```
f '(c) / g '(c) = [f(b) - f(a)] / [g(b) - g(a)]
```

Special Case: If we take g(x) = x, then g'(x) = 1, g(b) – g(a) = b – a, and Cauchy's theorem becomes LMVT! So LMVT is a special case of Cauchy's theorem.

7.2 Why is $g'(x) \neq 0$ Required?

If g'(x) = 0 at some point, the right side of the formula would involve dividing by zero (since g(b) – g(a) could be zero). To avoid this, we need $g'(x) \neq 0$ throughout (a, b).

Ex 7 Cauchy's MVT — Standard Example

Verify Cauchy's MVT for $f(x) = x^2$ and $g(x) = x^3$ on [1, 2].

STEP 1

$f(x) = x^2$, $g(x) = x^3$, $[a,b] = [1,2]$

STEP 3 – VERIFY CONDITIONS

Cond 1: Both are polynomials → continuous on [1,2]. ✓

Cond 2: Both are polynomials → differentiable on (1,2). ✓

Cond 3: $g'(x) = 3x^2$. For $x \in (1,2)$, $x > 0$, so $g'(x) = 3x^2 > 0 \neq 0$. ✓

STEP 4 – COMPUTE CAUCHY'S FORMULA RHS

$f(1) = 1$, $f(2) = 4$, so $f(2) - f(1) = 3$

$g(1) = 1$, $g(2) = 8$, so $g(2) - g(1) = 7$

$\text{RHS} = [f(2)-f(1)] / [g(2)-g(1)] = 3/7$

STEP 5 – FIND F'(X) AND G'(X)

$f(x) = x^2 \to f'(x) = 2x$

$g(x) = x^3 \to g'(x) = 3x^2$

STEP 6 – SET F'(C)/G'(C) = 3/7

$2c / (3c^2) = 3/7$

$2 / (3c) = 3/7$

Cross multiply: $14 = 9c$

$c = 14/9 \approx 1.556$

Check: Is $14/9 \in (1,2)$? Yes, since $1 < 1.556 < 2$. ✓

✅ **$c = 14/9 \in (1, 2)$, $f'(c)/g'(c) = 3/7$**

Ex 8 Cauchy's MVT — Exponential Functions

Verify Cauchy's MVT for $f(x) = e^x$, $g(x) = e^{-x}$ on $[0, 1]$.

STEP 3 – CONDITIONS

Both e^x and e^{-x} are continuous and differentiable everywhere. ✓

$g'(x) = -e^{-x} \neq 0$ for any x. ✓

STEP 4 – COMPUTE RHS

$f(0) = e^0 = 1$, $f(1) = e$

$g(0) = e^0 = 1$, $g(1) = e^{-1} = 1/e$

$[f(1)-f(0)] / [g(1)-g(0)] = (e-1) / (1/e - 1) = (e-1) / [(1-e)/e] = (e-1) \times e/(1-e) = -e$

STEP 5 – DERIVATIVES

$f'(x) = e^x$, $g'(x) = -e^{-x}$

STEP 6 – SOLVE F'(C)/G'(C) = –E

$e^c / (-e^{-c}) = -e$

$-e^c \times e^c = -e \times (-1)$ ... simplifying:

$-e^{2c} = -e$

$e^{2c} = e$

$2c = 1$

$c = 1/2$

Check: $0 < 1/2 < 1$. ✓

✅ **$c = 1/2 \in (0, 1)$**

8 Taylor's Theorem

Expanding Any Function as an Infinite Power Series

8.1 Statement of Taylor's Theorem (Without Proof)

TAYLOR

Taylor's Theorem:

If f(x) is a function that can be differentiated n times in the interval [a, a+h], then:

$$f(a+h) = f(a) + h \cdot f'(a) + (h^2/2!) \cdot f''(a) + (h^3/3!) \cdot f'''(a) + \dots + (h^n/n!) \cdot f^{(n)}(a) + R_n$$

where R_n is the **remainder term** (error term).

Lagrange's form of Remainder:
$R_n = h^{(n+1)} / (n+1)! \times f^{(n+1)}(a + \theta h)$, for some $\theta \in (0, 1)$

8.2 Taylor's Series in x (about x = a)

If we replace h = x – a in Taylor's theorem, we get the expansion of f(x) about the point x = a:

$$f(x) = f(a) + (x-a)f'(a) + [(x-a)^2/2!]f''(a) + [(x-a)^3/3!]f'''(a) + \dots$$

8.3 Understanding the Notation n!

n! (n factorial) means: n! = n × (n−1) × (n−2) × ... × 2 × 1

n	n!
0!	1 (by definition)
1!	1
2!	2
3!	6
4!	24
5!	120

8.4 Understanding $f^{(n)}$ Notation

$f^{(n)}(a)$ means the n-th derivative of f evaluated at x = a:

- $f^{(0)}(a) = f(a)$ — the function itself
- $f^{(1)}(a) = f'(a)$ — first derivative at a
- $f^{(2)}(a) = f''(a)$ — second derivative at a
- $f^{(3)}(a) = f'''(a)$ — third derivative at a

Ex 9 Taylor's Series — Expand sin x about x = 0

Expand $f(x) = \sin x$ using Taylor's series about $x = 0$ (up to x^5 term).

STEP 1

$f(x) = \sin x$, point of expansion $a = 0$

STEP 5 – FIND ALL DERIVATIVES AND THEIR VALUES AT X = 0

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	sin x	0
1	cos x	1
2	−sin x	0
3	−cos x	−1
4	sin x	0
5	cos x	1

Note: The derivatives of sin x repeat in a cycle of 4.

STEP 6 – SUBSTITUTE INTO TAYLOR'S FORMULA

$f(x) = f(0) + x \cdot f'(0) + (x^2/2!) \cdot f''(0) + (x^3/3!) \cdot f'''(0) + (x^4/4!) \cdot f^{(4)}(0) + (x^5/5!) \cdot f^{(5)}(0) + \ldots$

$= 0 + x \cdot (1) + (x^2/2!) \cdot (0) + (x^3/3!) \cdot (-1) + (x^4/4!) \cdot (0) + (x^5/5!) \cdot (1) + \ldots$

$= x - x^3/6 + x^5/120 - \ldots$

$\sin x = x - x^3/3! + x^5/5! - x^7/7! + \ldots = \Sigma (-1)^n x\text{^}(2n+1)/(2n+1)!$

Ex 10 Taylor's Expansion about $x = \pi/4$

Expand $f(x) = \sin x$ in powers of $(x - \pi/4)$ up to the third order term.

STEP 5 – DERIVATIVES AT A = Π/4

n	$f^{(n)}(x)$	$f^{(n)}(\pi/4)$
0	$\sin x$	$1/\sqrt{2}$
1	$\cos x$	$1/\sqrt{2}$
2	$-\sin x$	$-1/\sqrt{2}$
3	$-\cos x$	$-1/\sqrt{2}$

STEP 6 – APPLY TAYLOR'S FORMULA ABOUT A = Π/4

Let $h = x - \pi/4$

$\sin x = (1/\sqrt{2}) + (1/\sqrt{2})h - (1/\sqrt{2})(h^2/2!) - (1/\sqrt{2})(h^3/3!) + \ldots$

$= (1/\sqrt{2})[1 + (x-\pi/4) - (x-\pi/4)^2/2 - (x-\pi/4)^3/6 + \ldots]$

$\sin x = (1/\sqrt{2})[1 + (x-\pi/4) - (x-\pi/4)^2/2 - (x-\pi/4)^3/6 + \ldots]$

9 Maclaurin's Theorem

Taylor's Theorem at a = 0 — The Most Commonly Used Form

9.1 Statement

MACLAURIN

Maclaurin's Theorem:

Maclaurin's theorem is a **special case of Taylor's theorem** where the expansion is about **$x = 0$** (i.e., $a = 0$):

$$f(x) = f(0) + x \cdot f'(0) + (x^2/2!) \cdot f''(0) + (x^3/3!) \cdot f'''(0) + \ldots + (x^n/n!) \cdot f^{(n)}(0) + R_n$$

where $R_n = x^{(n+1)}/(n+1)! \times f^{(n+1)}(\theta x)$ for some $\theta \in (0,1)$

How to remember: Maclaurin = Taylor about the ORIGIN ($x = 0$). The term "about $x = 0$" simply means: substitute $a = 0$ and $h = x$ into Taylor's formula.

9.2 Method to Expand Using Maclaurin's Theorem

1. Find $f(0)$, $f'(0)$, $f''(0)$, $f'''(0)$, ... by computing each derivative and substituting $x = 0$
2. Substitute into the formula: $f(x) = f(0) + xf'(0) + x^2/2!\, f''(0) + \ldots$
3. Write as many terms as required

Ex 11 Maclaurin's Series — e^x

Expand $f(x) = e^x$ using Maclaurin's series.

STEP 5 – FIND ALL DERIVATIVES AND VALUES AT X = 0

n	$f^{(n)}(x)$	$f^{(n)}(0)$	Explanation
0	e^x	$e^0 = 1$	Function itself
1	e^x	1	Derivative of e^x is e^x (special!)
2	e^x	1	Second derivative also e^x
3	e^x	1	All derivatives of e^x are e^x
n	e^x	1	$f^{(n)}(0) = 1$ for all n

STEP 6 – SUBSTITUTE INTO MACLAURIN'S FORMULA

$f(x) = f(0) + x \cdot f'(0) + (x^2/2!) \cdot f''(0) + (x^3/3!) \cdot f'''(0) + \ldots$

$= 1 + x \cdot (1) + (x^2/2!) \cdot (1) + (x^3/3!) \cdot (1) + (x^4/4!) \cdot (1) + \ldots$

$= 1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots = \Sigma\, x^n/n! \text{ (for all } x\text{)}$$

💡 **Key Fact:** e^x is special — all its derivatives are e^x itself, so all coefficients in the expansion equal $1/(n!)$. This is one of the most beautiful results in mathematics!

Ex 12 Maclaurin's Series — cos x

Expand $f(x) = \cos x$ using Maclaurin's series.

STEP 5 – DERIVATIVES AT X = 0

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\cos x$	1
1	$-\sin x$	0
2	$-\cos x$	-1
3	$\sin x$	0
4	$\cos x$	1
5	$-\sin x$	0

STEP 6 – BUILD THE SERIES

$\cos x = 1 + x(0) + (x^2/2!)(-1) + (x^3/3!)(0) + (x^4/4!)(1) + \ldots$

$= 1 - x^2/2! + x^4/4! - x^6/6! + \ldots$

$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \ldots = \Sigma\, (-1)^n\, x^{(2n)}/(2n)!$

Ex 13 Maclaurin's Series — log(1+x)

Expand $f(x) = \log(1 + x)$ using Maclaurin's series. State the range of validity.

STEP 5 – DERIVATIVES AND VALUES AT X = 0

$f(x) = \log(1+x) \rightarrow f(0) = \log(1) = 0$

$f'(x) = 1/(1+x) \rightarrow f'(0) = 1$

$f''(x) = -1/(1+x)^2 \rightarrow f''(0) = -1$

$f'''(x) = 2/(1+x)^3 \rightarrow f'''(0) = 2$

$f^{(4)}(x) = -6/(1+x)^4 \rightarrow f^{(4)}(0) = -6$

$f^{(n)}(0) = (-1)^{(n-1)} \times (n-1)!$ for $n \geq 1$

STEP 6 – BUILD THE SERIES

$\log(1+x) = 0 + x(1) + (x^2/2!)(-1) + (x^3/3!)(2) + (x^4/4!)(-6) + \ldots$

$= x - x^2/2 + 2x^3/6 - 6x^4/24 + \ldots$

$= x - x^2/2 + x^3/3 - x^4/4 + \ldots$

$\log(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \ldots$ (valid for $-1 < x \leq 1$)

10 Standard Expansions

All Important Series You Must Memorize

10.1 Complete Table of Standard Maclaurin Series

Function	Expansion	Range of Validity
e^x	$1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$	All real x $(-\infty, \infty)$
$e^{(-x)}$	$1 - x + x^2/2! - x^3/3! + x^4/4! - \ldots$	All real x
sin x	$x - x^3/3! + x^5/5! - x^7/7! + \ldots$	All real x
cos x	$1 - x^2/2! + x^4/4! - x^6/6! + \ldots$	All real x
tan x	$x + x^3/3 + 2x^5/15 + \ldots$	$\|x\| < \pi/2$
log(1+x)	$x - x^2/2 + x^3/3 - x^4/4 + \ldots$	$-1 < x \leq 1$
log(1−x)	$-x - x^2/2 - x^3/3 - x^4/4 - \ldots$	$-1 \leq x < 1$
$(1+x)^n$	$1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \ldots$	$\|x\| < 1$
sinh x	$x + x^3/3! + x^5/5! + x^7/7! + \ldots$	All real x
cosh x	$1 + x^2/2! + x^4/4! + x^6/6! + \ldots$	All real x

10.2 Memory Tricks for Standard Series

sin x vs cos x

- **sin x:** Starts with x (odd powers, alternating signs: $+x$, $-x^3$, $+x^5$...)
- **cos x:** Starts with 1 (even powers, alternating signs: $+1$, $-x^2$, $+x^4$...)
- Think: sin = Sine starts with x, cos = Constant starts first!

e^x vs log(1+x)

- **e^x:** All positive, all factorials in denominator
- **log(1+x):** No factorials, denominators are 1,2,3,4... alternating signs

Ex 14 Maclaurin's Series — tan x (Up to x^3)

Expand $f(x) = \tan x$ using Maclaurin's series up to x^3 term.

STEP 5 – DERIVATIVES AT X = 0

$f(x) = \tan x \rightarrow f(0) = 0$

$f'(x) = \sec^2 x \rightarrow f'(0) = 1$

$f''(x) = 2\sec^2 x \cdot \tan x \rightarrow f''(0) = 2(1)(0) = 0$

$f'''(x) = 2[2\sec^2 x \cdot \tan x \cdot \tan x + \sec^2 x \cdot \sec^2 x] = 2[2\sec^2 x \cdot \tan^2 x + \sec^4 x]$

At $x=0$: $f'''(0) = 2[2(1)(0) + 1] = 2$

STEP 6 – SUBSTITUTION

$\tan x = 0 + x(1) + (x^2/2)(0) + (x^3/6)(2) + \ldots$

$= x + x^3/3 + \ldots$

$\tan x = x + x^3/3 + 2x^5/15 + \ldots$

11

Applications of Taylor & Maclaurin Series

Solving Real Problems Using Series Expansions

11.1 Why Are These Expansions Useful?

Series expansions are used to:

- **Compute approximate values** of functions like sin(1°), e^0.1, log(1.05)
- **Evaluate limits** involving indeterminate forms (0/0)
- **Integrate complicated functions** by expanding first
- **Solve differential equations** approximately

Ex 15 Approximation — Compute e^0.1 Approximately

Using Maclaurin's series, find an approximate value of e^0.1 (correct to 4 decimal places).

STEP 1 – IDENTIFY

We need to find e^x at x = 0.1. We use the Maclaurin series for e^x.

STEP 4 – WRITE THE SERIES

$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$

STEP 5 – SUBSTITUTE X = 0.1

$e^{(0.1)} = 1 + (0.1) + (0.1)^2/2! + (0.1)^3/3! + (0.1)^4/4! + \ldots$

$= 1 + 0.1 + 0.01/2 + 0.001/6 + 0.0001/24 + \ldots$

$= 1 + 0.1 + 0.005 + 0.0001667 + 0.0000042 + \ldots$

STEP 6 – ADD THE TERMS

$e^{(0.1)} \approx 1 + 0.1 + 0.005 + 0.000167 + 0.0000042$

≈ 1.10517 (to 5 decimal places)

(Actual value: $e^{0.1} = 1.10517\ldots$ ✓)

$e^{0.1} \approx 1.1052$ (4 decimal places)

💡 **Note:** We only needed a few terms because x = 0.1 is small. The series converges quickly for small x.

Ex 16 Approximation — Find sin(31°)

Find the approximate value of sin 31° using Taylor's expansion of sin x about $x = 30° = \pi/6$.

STEP 1 – CONVERT TO RADIANS

$31° = 31 \times \pi/180$ radians

Let $a = 30° = \pi/6$, $h = 1° = \pi/180$ (small increment)

We expand $\sin(\pi/6 + \pi/180)$ using Taylor's series.

STEP 5 – DERIVATIVES AT A = Π/6

$f(a) = \sin(\pi/6) = 1/2 = 0.5$

$f'(a) = \cos(\pi/6) = \sqrt{3}/2 \approx 0.8660$

$f''(a) = -\sin(\pi/6) = -0.5$

STEP 6 – TAYLOR'S EXPANSION

$h = \pi/180 \approx 0.01745$

$\sin(31°) = f(a) + h \cdot f'(a) + (h^2/2!) \cdot f''(a) + \ldots$

$= 0.5 + (0.01745)(0.8660) + (0.01745^2/2)(-0.5) + \ldots$

$= 0.5 + 0.015112 - 0.000076 + \ldots$

≈ 0.51504

$\sin 31° \approx 0.5150$ (Actual: 0.5150 ✓)

Ex 17 Evaluate Limit Using Series — $\lim_{x\to 0} \sin(x)/x$

Evaluate $\lim_{x\to 0} [\sin x / x]$ using Maclaurin's series.

STEP 4 – EXPAND SIN X

$\sin x = x - x^3/3! + x^5/5! - \dots$

$\sin x / x = (x - x^3/6 + x^5/120 - \dots) / x$

$= 1 - x^2/6 + x^4/120 - \dots$

STEP 6 – TAKE THE LIMIT

$\lim_{x\to 0} [\sin x / x] = \lim_{x\to 0} [1 - x^2/6 + x^4/120 - \dots]$

As $x \to 0$, all terms with x vanish:

$= 1 - 0 + 0 - \dots = \mathbf{1}$

$\lim_{x\to 0} \sin(x)/x = 1$

12 Approximation Problems

Numerical Approximations Using Series Expansions

Ex 18 Approximate log(1.1)

Find log(1.1) using Maclaurin's expansion of log(1+x), up to x^4 term.

STEP 4 – SERIES

$\log(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \ldots$

STEP 5 – SUBSTITUTE X = 0.1

$\log(1.1) = 0.1 - (0.1)^2/2 + (0.1)^3/3 - (0.1)^4/4 + \ldots$

$= 0.1 - 0.01/2 + 0.001/3 - 0.0001/4 + \ldots$

$= 0.1 - 0.005 + 0.000333 - 0.000025 + \ldots$

≈ 0.09531

log(1.1) ≈ 0.0953 (Actual: 0.09531 ✓)

Ex 19 Approximate cos(1°)

Find cos(1°) approximately using the series for cos x. (1° = π/180 radians)

STEP 5 – SUBSTITUTE X = Π/180 ≈ 0.01745

$\cos x = 1 - x^2/2! + x^4/4! - \ldots$

$\cos(1^\circ) = 1 - (0.01745)^2/2 + \ldots$

$= 1 - 0.000304/2 +$ (negligible)

$= 1 - 0.000152 \approx 0.99985$

cos(1°) ≈ 0.9998 (Actual: 0.99985 ✓)

Ex 20 Approximate $\sqrt{1.02}$

Using the binomial expansion $(1+x)^{1/2}$, find $\sqrt{1.02}$ approximately.

STEP 4 – BINOMIAL SERIES

$(1+x)^n = 1 + nx + n(n-1)/2! \times x^2 + \ldots$

Here $n = 1/2$, $x = 0.02$ (small!)

STEP 5 – SUBSTITUTE

$\sqrt{1.02} = (1+0.02)^{1/2}$

$= 1 + (1/2)(0.02) + (1/2)(-1/2)/2! \times (0.02)^2 + \ldots$

$= 1 + 0.01 - (1/8)(0.0004) + \ldots$

$= 1 + 0.01 - 0.00005 + \ldots$

≈ 1.00995

$\sqrt{1.02} \approx 1.0100$ (Actual: 1.00995 ✓)

13

Error Estimation

How Accurate Is Our Approximation?

13.1 The Remainder Term

When we truncate (cut off) an infinite series after n terms, there is an **error**. This error is given by the remainder term R_n.

$$|R_n| \leq M \cdot |x|^{(n+1)} / (n+1)!$$

where M = maximum value of $|f^{(n+1)}(x)|$ on the interval

How to use this: If you want to find the error when computing $e^{0.5}$ using 4 terms, find the 5th derivative bound and plug into the formula.

Ex 21 **Error Estimation for e^x**

Estimate the error when $e^{0.5}$ is approximated by the first 4 terms of its Maclaurin series.

STEP 4 – THE 4-TERM APPROXIMATION

$e^x \approx 1 + x + x^2/2! + x^3/3!$ (first 4 terms means n = 3)

The next term is $x^4/4!$ (the error term is R_3)

STEP 5 – FIND THE BOUND FOR $F^{(4)}(X)$

$f(x) = e^x \rightarrow$ all derivatives $= e^x$

On [0, 0.5], the maximum value of e^x is $e^{(0.5)} \approx 1.649$

So $M = e^{(0.5)} \approx 1.649$

STEP 6 – APPLY ERROR FORMULA

$|R_3| \leq M \cdot |x|^4 / 4! = 1.649 \times (0.5)^4 / 24$

$= 1.649 \times 0.0625 / 24$

$= 0.1031 / 24$

≈ 0.0043

Maximum error ≤ 0.0043 (i.e., the approximation is accurate to within 0.0043)

14 Solved Examples — Complete Bank

40+ Ultra-Detailed Step-by-Step Solutions

Ex 22 Rolle's — $f(x) = x(x-1)(x-2)$ on $[0,2]$

Verify Rolle's Theorem for $f(x) = x(x-1)(x-2)$ on $[0, 2]$.

STEP 1

$f(x) = x(x-1)(x-2) = x^3 - 3x^2 + 2x$, $[0, 2]$

STEP 3 – CONDITIONS

Polynomial → Continuous on $[0,2]$ ✓, Differentiable on $(0,2)$ ✓

$f(0) = 0\cdot(-1)\cdot(-2) = 0$

$f(2) = 2\cdot1\cdot0 = 0$

$f(0) = f(2) = 0$ ✓

STEP 5 – DIFFERENTIATE

$f(x) = x^3 - 3x^2 + 2x$

$f'(x) = 3x^2 - 6x + 2$

STEP 6 – SOLVE F'(C) = 0

$3c^2 - 6c + 2 = 0$

$c = [6 \pm \sqrt{(36-24)}] / 6 = [6 \pm \sqrt{12}] / 6 = [6 \pm 2\sqrt{3}] / 6 = 1 \pm 1/\sqrt{3}$

$c_1 = 1 - 1/\sqrt{3} \approx 0.423$, $c_2 = 1 + 1/\sqrt{3} \approx 1.577$

Both lie in $(0, 2)$. ✓

✅ **$c = 1 \pm 1/\sqrt{3} \in (0, 2)$**

Ex 23 Rolle's — $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$

Verify Rolle's Theorem for $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$.

STEP 3 – CONDITIONS

e^x, $\sin x$, $\cos x$ are all continuous and differentiable everywhere.

Product of continuous/differentiable functions is continuous/differentiable. ✓

$f(\pi/4) = e^{(\pi/4)}(\sin(\pi/4) - \cos(\pi/4)) = e^{(\pi/4)}(1/\sqrt{2} - 1/\sqrt{2}) = 0$

$f(5\pi/4) = e^{(5\pi/4)}(\sin(5\pi/4) - \cos(5\pi/4)) = e^{(5\pi/4)}(-1/\sqrt{2} - (-1/\sqrt{2})) = e^{(5\pi/4)}(0) = 0$

$f(\pi/4) = f(5\pi/4) = 0$ ✓

STEP 5 – FIND F'(X) USING PRODUCT RULE

$f(x) = e^x \cdot (\sin x - \cos x)$

Product rule: $d/dx[u \cdot v] = u'v + uv'$

Let $u = e^x \rightarrow u' = e^x$

Let $v = \sin x - \cos x \rightarrow v' = \cos x + \sin x$

$f'(x) = e^x(\sin x - \cos x) + e^x(\cos x + \sin x)$

$= e^x[\sin x - \cos x + \cos x + \sin x]$

$= e^x[2 \sin x]$

$= 2e^x \sin x$

STEP 6 – SET F'(C) = 0

$2e^c \sin c = 0$

Since $e^c > 0$ always, we need $\sin c = 0$

$\sin c = 0 \rightarrow c = n\pi$

In $(\pi/4, 5\pi/4)$: $c = \pi \approx 3.14$

$\pi/4 \approx 0.785 < \pi < 5\pi/4 \approx 3.927$ ✓

✅ $c = \pi \in (\pi/4, 5\pi/4)$

Ex 24 LMVT — $f(x) = \sin x$ on $[0, \pi/2]$

Find the value of c guaranteed by LMVT for $f(x) = \sin x$ on $[0, \pi/2]$.

STEP 4 – LMVT CALCULATION

$f(0) = \sin 0 = 0$, $f(\pi/2) = \sin(\pi/2) = 1$

$[f(\pi/2)-f(0)] / (\pi/2-0) = 1/(\pi/2) = 2/\pi$

STEP 5 – DIFFERENTIATE

$f'(x) = \cos x$

STEP 6 – SET F'(C) = 2/Π

$\cos c = 2/\pi \approx 0.6366$

$c = \cos^{-1}(2/\pi) \approx 0.8807$ radians

Check: $0 < 0.8807 < \pi/2 \approx 1.5708$. ✓

$c = \cos^{-1}(2/\pi) \approx 0.881 \in (0, \pi/2)$

Ex 25 Application of LMVT — Prove Inequality

Using LMVT, prove that $|\sin a - \sin b| \leq |a - b|$ for all real a, b.

STEP 2 – IDENTIFY APPROACH

Apply LMVT to $f(x) = \sin x$ on the interval $[a, b]$ (assuming $a < b$).

$f(x) = \sin x$ is continuous and differentiable everywhere.

STEP 4 – APPLY LMVT

By LMVT, $\exists\, c \in (a,b)$ such that:

$f'(c) = [f(b) - f(a)] / (b - a)$

$\cos c = (\sin b - \sin a) / (b - a)$

STEP 6 – USE THE BOUND $|\cos c| \leq 1$

$|(\sin b - \sin a) / (b - a)| = |\cos c| \leq 1$

Therefore: $|\sin b - \sin a| \leq |b - a|$

Which is $|\sin a - \sin b| \leq |a - b|$ (since absolute value is symmetric). ■

Proved: $|\sin a - \sin b| \leq |a - b|$

Ex 26 Cauchy's MVT — Trig Functions

Verify Cauchy's MVT for $f(x) = \sin x$, $g(x) = \cos x$ on $[0, \pi/2]$.

STEP 3 – CONDITIONS

Both $\sin x$ and $\cos x$ continuous and differentiable everywhere. ✓

$g'(x) = -\sin x$. For $x \in (0, \pi/2)$, $\sin x > 0$, so $g'(x) \neq 0$. ✓

STEP 4 – RHS

$f(0) = 0$, $f(\pi/2) = 1 \rightarrow f(\pi/2) - f(0) = 1$

$g(0) = 1$, $g(\pi/2) = 0 \rightarrow g(\pi/2) - g(0) = -1$

RHS $= 1/(-1) = -1$

STEP 5 – DERIVATIVES

$f'(x) = \cos x$, $g'(x) = -\sin x$

STEP 6 – SET F'(C)/G'(C) = −1

$\cos c \,/\, (-\sin c) = -1$

$-\cot c = -1$

$\cot c = 1$

$\tan c = 1$

$c = \pi/4$

Check: $0 < \pi/4 < \pi/2$. ✓

✅ **$c = \pi/4 \in (0, \pi/2)$**

Ex 27 Taylor's Series of e^x about x = 1

Expand e^x in powers of $(x-1)$ using Taylor's series up to the 4th degree term.

STEP 5 – DERIVATIVES AT A = 1

All derivatives of e^x equal e^x.

So $f^{(n)}(1) = e^1 = e$ for all $n \geq 0$.

STEP 6 – TAYLOR'S SERIES

$e^x = e + e(x-1) + e(x-1)^2/2! + e(x-1)^3/3! + e(x-1)^4/4! + \ldots$

$= e[1 + (x-1) + (x-1)^2/2 + (x-1)^3/6 + (x-1)^4/24 + \ldots]$

$$e^x = e\cdot\Sigma\, (x-1)^n/n! \text{ for } n = 0,1,2,3,\ldots$$

Ex 28 Maclaurin's Series — (1+x)^n (Binomial)

Expand $(1+x)^n$ using Maclaurin's series.

STEP 5 – DERIVATIVES AT X = 0

$f(x) = (1+x)^n \rightarrow f(0) = 1$

$f'(x) = n(1+x)^{(n-1)} \rightarrow f'(0) = n$

$f''(x) = n(n-1)(1+x)^{(n-2)} \rightarrow f''(0) = n(n-1)$

$f'''(x) = n(n-1)(n-2)(1+x)^{(n-3)} \rightarrow f'''(0) = n(n-1)(n-2)$

$f^{(r)}(0) = n(n-1)(n-2)\ldots(n-r+1)$

STEP 6 – SUBSTITUTION

$(1+x)^n = 1 + nx + n(n-1)/2! \times x^2 + n(n-1)(n-2)/3! \times x^3 + \ldots$

This is the **Binomial Series**.

$$(1+x)^n = 1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \ldots$$

💡 **Special cases:**

$n = 1/2$: $(1+x)^{(1/2)} = \sqrt{(1+x)} = 1 + x/2 - x^2/8 + \ldots$

$n = -1$: $1/(1+x) = 1 - x + x^2 - x^3 + \ldots$

$n = 2$: $(1+x)^2 = 1 + 2x + x^2$ (exact — series terminates!)

Ex 29 Taylor's Expansion of log x about x = 1

Expand log x in powers of (x−1) up to the 4th degree using Taylor's series.

STEP 5 – DERIVATIVES AT A = 1

$f(x) = \log x \to f(1) = 0$

$f'(x) = 1/x \to f'(1) = 1$

$f''(x) = -1/x^2 \to f''(1) = -1$

$f'''(x) = 2/x^3 \to f'''(1) = 2$

$f^{(4)}(x) = -6/x^4 \to f^{(4)}(1) = -6$

STEP 6 – TAYLOR'S FORMULA

$\log x = 0 + (x-1)(1) + (x-1)^2/2!(-1) + (x-1)^3/3!(2) + (x-1)^4/4!(-6) + \ldots$

$= (x-1) - (x-1)^2/2 + (x-1)^3/3 - (x-1)^4/4 + \ldots$

$$\log x = (x-1) - (x-1)^2/2 + (x-1)^3/3 - (x-1)^4/4 + \ldots$$

Ex 30 Rolle's Theorem — $f(x) = \sin x \cdot e^{-x}$ on $[0, \pi]$

Verify Rolle's theorem for $f(x) = e^{-x} \sin x$ on $[0, \pi]$.

STEP 3 – CONDITIONS

Product of continuous/differentiable functions → f is continuous and differentiable. ✓

$f(0) = e^0 \sin 0 = 1 \cdot 0 = 0$

$f(\pi) = e^{-\pi} \sin \pi = e^{-\pi} \cdot 0 = 0$

$f(0) = f(\pi) = 0$ ✓

STEP 5 – DIFFERENTIATE USING PRODUCT RULE

$f(x) = e^{-x} \sin x$

$u = e^{-x}$, $u' = -e^{-x}$

$v = \sin x$, $v' = \cos x$

$f'(x) = -e^{-x} \sin x + e^{-x} \cos x = e^{-x}(\cos x - \sin x)$

STEP 6 – SOLVE F'(C) = 0

$e^{-c}(\cos c - \sin c) = 0$

Since $e^{-c} > 0$ always: $\cos c - \sin c = 0$

$\cos c = \sin c \rightarrow \tan c = 1 \rightarrow c = \pi/4$

Check: $0 < \pi/4 < \pi$ ✓

✅ **$c = \pi/4 \in (0, \pi)$**

Ex 31 Expansion of $\sin^2 x$ using Maclaurin's Series

Expand $f(x) = \sin^2 x$ using Maclaurin's series up to x^6 term.

STEP 2 – IDENTITY TRICK

Use the identity: $\sin^2 x = (1 - \cos 2x) / 2$

This avoids repeatedly differentiating $\sin^2 x$ (which gets complicated).

STEP 4 – USE STANDARD SERIES FOR COS 2X

$\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \ldots$

Replace x with 2x:

$\cos 2x = 1 - (2x)^2/2! + (2x)^4/4! - (2x)^6/6! + \ldots$

$= 1 - 4x^2/2 + 16x^4/24 - 64x^6/720 + \ldots$

$= 1 - 2x^2 + 2x^4/3 - 4x^6/45 + \ldots$

STEP 6 – COMPUTE SIN²X

$\sin^2 x = (1 - \cos 2x) / 2$

$= (1 - [1 - 2x^2 + 2x^4/3 - 4x^6/45 + \ldots]) / 2$

$= (2x^2 - 2x^4/3 + 4x^6/45 - \ldots) / 2$

$= x^2 - x^4/3 + 2x^6/45 - \ldots$

$\sin^2 x = x^2 - x^4/3 + 2x^6/45 - \ldots$

💡 **Trick:** Always look for trig identities to simplify before expanding. It saves you from heavy differentiation!

Ex 32 Rolle's — Checking if Theorem is Applicable

Can Rolle's Theorem be applied to $f(x) = |x|$ on $[-1, 1]$? Justify.

STEP 3 – CHECK ALL CONDITIONS

Condition 1 (Continuity): $|x|$ is continuous on $[-1, 1]$. ✓

Condition 2 (Differentiability):

$f(x) = |x| = x$ if $x \geq 0$, $= -x$ if $x < 0$

At $x = 0$: Left derivative $= \lim(h\to 0^-) [|0+h|-|0|]/h = \lim [|h|/h] = \lim [-h/h] = -1$

Right derivative $= \lim(h\to 0^+) [|h|/h] = \lim [h/h] = 1$

Left ≠ Right → f is **NOT differentiable at $x = 0$**. ✗

Condition 2 FAILS!

Condition 3: $f(-1) = 1$, $f(1) = 1$, so $f(-1) = f(1)$. ✓

STEP 8 – CONCLUSION

Since Condition 2 (differentiability) fails at $x = 0 \in (-1, 1)$, **Rolle's Theorem is NOT applicable** to $f(x) = |x|$ on $[-1, 1]$.

❌ Rolle's Theorem is NOT applicable — $f(x) = |x|$ is not differentiable at $x = 0$.

💡 **Important:** This is a classic example showing that all THREE conditions must be verified. Just because $f(a) = f(b)$ doesn't guarantee Rolle's theorem applies!

Ex 33 LMVT — Prove $e^x > 1 + x$ for $x > 0$

Using LMVT, prove that $e^x > 1 + x$ for all $x > 0$.

STEP 2 – SETUP

Consider $f(t) = e^t$ on $[0, x]$ where $x > 0$.

f is continuous and differentiable. Apply LMVT.

STEP 4 – APPLY LMVT

There exists $c \in (0, x)$ such that:

$f'(c) = [f(x) - f(0)] / (x - 0)$

$e^c = (e^x - 1) / x$

STEP 6 – USE C ∈ (0, X)

Since $c > 0$, $e^c > e^0 = 1$ (exponential is increasing)

So: $(e^x - 1) / x = e^c > 1$

Therefore: $e^x - 1 > x$ (multiplying both sides by $x > 0$)

Hence: $e^x > 1 + x$ ■

Proved: $e^x > 1 + x$ for all $x > 0$

Ex 34 Expansion of $e^{(\sin x)}$ up to x^4

Expand $e^{(\sin x)}$ using Maclaurin's series up to x^4 terms.

STEP 2 – APPROACH

Replace x by sin x in the expansion of e^x.

$\sin x = x - x^3/6 + \dots$

Let $u = \sin x = x - x^3/6 + \dots$ (keeping only terms up to x^4)

STEP 4 – SUBSTITUTE INTO $E^U = 1 + U + U^2/2 + U^3/6 + \dots$

$u = x - x^3/6$

$u^2 = (x - x^3/6)^2 = x^2 - 2x \cdot x^3/6 + \dots = x^2 - x^4/3 + \dots$ (keep up to x^4)

$u^3 = x^3 + \dots$ (higher order)

$e^{(\sin x)} = 1 + u + u^2/2 + u^3/6 + \dots$

$= 1 + (x - x^3/6) + (x^2 - x^4/3)/2 + x^3/6 + \dots$

$= 1 + x + x^2/2 - x^3/6 + x^3/6 - x^4/6 + \dots$

$= 1 + x + x^2/2 + 0 \cdot x^3 - x^4/8 + \dots$

$$e^{(\sin x)} = 1 + x + x^2/2 - x^4/8 + \dots$$

Ex 35 Maclaurin Series — (1+x)log(1+x)

Expand (1+x)log(1+x) using Maclaurin's series up to x^4.

STEP 4 – USE KNOWN SERIES

$\log(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \dots$

$(1+x) \times \log(1+x) = (1+x)(x - x^2/2 + x^3/3 - x^4/4 + \dots)$

STEP 5 – MULTIPLY TERM BY TERM

$= 1 \cdot (x - x^2/2 + x^3/3 - x^4/4) + x \cdot (x - x^2/2 + x^3/3 - x^4/4) + \dots$

$= (x - x^2/2 + x^3/3 - x^4/4) + (x^2 - x^3/2 + x^4/3) + \dots$

$= x + (-1/2+1)x^2 + (1/3-1/2)x^3 + (-1/4+1/3)x^4 + \dots$

$= x + x^2/2 - x^3/6 + x^4/12 + \dots$

$$(1+x)\log(1+x) = x + x^2/2 - x^3/6 + x^4/12 - \dots$$

Ex 36 Rolle's Theorem — Rational Function

Verify Rolle's Theorem for $f(x) = (x^2 - 1)(x - 2)$ on $[-1, 1]$.

STEP 3 – CONDITIONS

Polynomial → Continuous on $[-1,1]$ and Differentiable on $(-1,1)$. ✓

$f(-1) = (1-1)(-1-2) = 0\cdot(-3) = 0$

$f(1) = (1-1)(1-2) = 0\cdot(-1) = 0$

$f(-1) = f(1) = 0$ ✓

STEP 5 – EXPAND AND DIFFERENTIATE

$f(x) = (x^2-1)(x-2) = x^3 - 2x^2 - x + 2$

$f'(x) = 3x^2 - 4x - 1$

STEP 6 – SOLVE

$3c^2 - 4c - 1 = 0$

$c = [4 \pm \sqrt{(16+12)}] / 6 = [4 \pm \sqrt{28}] / 6 = [4 \pm 2\sqrt{7}] / 6 = (2 \pm \sqrt{7})/3$

$c_1 = (2 - \sqrt{7})/3 \approx (2-2.646)/3 \approx -0.215$

$c_2 = (2 + \sqrt{7})/3 \approx (2+2.646)/3 \approx 1.549$ (not in $(-1,1)$)

$c_1 \approx -0.215 \in (-1, 1)$ ✓

✅ **$c = (2-\sqrt{7})/3 \approx -0.215 \in (-1, 1)$**

Ex 37 LMVT — $f(x) = \sqrt{x}$ on [1, 4]

Verify LMVT for $f(x) = \sqrt{x}$ on [1, 4].

STEP 3 – CONDITIONS

$\sqrt{x}$ is continuous on [1,4] ✓ (positive interval, no issues)

$f'(x) = 1/(2\sqrt{x})$ exists for $x \in (1,4)$ ✓

STEP 4 – COMPUTE AVERAGE RATE

$f(1) = 1$, $f(4) = 2$

$[f(4)-f(1)]/(4-1) = (2-1)/3 = 1/3$

STEP 6 – SOLVE F'(C) = 1/3

$1/(2\sqrt{c}) = 1/3$

$2\sqrt{c} = 3$

$\sqrt{c} = 3/2$

$c = 9/4 = 2.25$

Check: $1 < 2.25 < 4$ ✓

✅ **$c = 9/4 = 2.25 \in (1, 4)$**

Ex 38 Maclaurin — Expand $e^x \cos x$ up to x^4

Expand $f(x) = e^x \cos x$ using Maclaurin's series up to x^4.

STEP 4 – MULTIPLY KNOWN SERIES

$e^x = 1 + x + x^2/2 + x^3/6 + x^4/24 + \ldots$

$\cos x = 1 - x^2/2 + x^4/24 - \ldots$

Product (collecting terms up to x^4):

$= (1 + x + x^2/2 + x^3/6 + x^4/24)(1 - x^2/2 + x^4/24)$

x^0: 1

x^1: x

x^2: $x^2/2 - x^2/2 = 0$

x^3: $x^3/6 - x^3/2 = x^3(1/6 - 1/2) = -x^3/3$

x^4: $x^4/24 - x^4/4 + x^4/24 = x^4(1/24 - 1/4 + 1/24) = x^4(2/24 - 6/24) = -x^4/6$

$e^x \cos x = 1 + x - x^3/3 - x^4/6 + \ldots$

Ex 39 Taylor's Expansion — $\cos x$ about $x = \pi/3$

Expand $\cos x$ in powers of $(x - \pi/3)$ up to the third degree.

STEP 5 – DERIVATIVES AT A = Π/3

$f(x) = \cos x \to f(\pi/3) = \cos(\pi/3) = 1/2$

$f'(x) = -\sin x \to f'(\pi/3) = -\sin(\pi/3) = -\sqrt{3}/2$

$f''(x) = -\cos x \to f''(\pi/3) = -1/2$

$f'''(x) = \sin x \to f'''(\pi/3) = \sqrt{3}/2$

STEP 6 – TAYLOR'S SERIES

$\cos x = 1/2 - (\sqrt{3}/2)(x-\pi/3) - (1/2)(x-\pi/3)^2/2! + (\sqrt{3}/2)(x-\pi/3)^3/3! + \ldots$

$= 1/2 - (\sqrt{3}/2)(x-\pi/3) - (x-\pi/3)^2/4 + (\sqrt{3}/12)(x-\pi/3)^3 + \ldots$

$\cos x = 1/2 - (\sqrt{3}/2)(x-\pi/3) - (x-\pi/3)^2/4 + (\sqrt{3}/12)(x-\pi/3)^3 + \ldots$

Ex 40 Evaluate $\lim_{x\to 0} [e^x - 1 - x] / x^2$

Using Maclaurin's series, evaluate $\lim_{x\to 0} [e^x - 1 - x] / x^2$.

STEP 4 – EXPAND NUMERATOR

$e^x = 1 + x + x^2/2! + x^3/3! + \ldots$

$e^x - 1 - x = x^2/2 + x^3/6 + x^4/24 + \ldots$

STEP 5 – DIVIDE BY X²

$[e^x - 1 - x] / x^2 = (x^2/2 + x^3/6 + \ldots) / x^2$

$= 1/2 + x/6 + x^2/24 + \ldots$

STEP 6 – TAKE LIMIT

$\lim_{x\to 0} [1/2 + x/6 + \ldots] = 1/2$

$\lim_{x\to 0} [e^x - 1 - x] / x^2 = 1/2$

Ex 41 Rolle's Theorem for f(x) = sin x + cos x on [0, 2π]

Verify Rolle's theorem for $f(x) = \sin x + \cos x$ on $[0, 2\pi]$.

STEP 3 – CONDITIONS

sin x + cos x is continuous and differentiable everywhere. ✓

$f(0) = \sin 0 + \cos 0 = 0 + 1 = 1$

$f(2\pi) = \sin 2\pi + \cos 2\pi = 0 + 1 = 1$

$f(0) = f(2\pi) = 1$ ✓

STEP 5 – DIFFERENTIATE

$f'(x) = \cos x - \sin x$

STEP 6 – SOLVE F'(C) = 0

$\cos c - \sin c = 0 \to \tan c = 1 \to c = \pi/4$ or $c = 5\pi/4$

Both $\pi/4$ and $5\pi/4$ lie in $(0, 2\pi)$. ✓

✅ $c = \pi/4$ and $c = 5\pi/4$ both $\in (0, 2\pi)$

Ex 42 Expansion of $\log(1+\sin x)$ up to x^4

Expand $\log(1 + \sin x)$ using Maclaurin's series up to x^4 term.

STEP 4 – SUBSTITUTE SIN X INTO LOG(1+U)

Let $u = \sin x = x - x^3/6 + \ldots$

$\log(1+u) = u - u^2/2 + u^3/3 - u^4/4 + \ldots$

$u = x - x^3/6$

$u^2 = x^2 - x^4/3 + \ldots$ (up to x^4)

$u^3 = x^3 + \ldots$ (keep x^3 only)

$u^4 = x^4 + \ldots$

STEP 5 – COLLECT TERMS

$\log(1+\sin x) = (x - x^3/6) - (x^2 - x^4/3)/2 + x^3/3 - x^4/4 + \ldots$

$= x - x^2/2 - x^3/6 + x^3/3 + x^4/6 - x^4/4 + \ldots$

$= x - x^2/2 + x^3/6 - x^4/12 + \ldots$

$$\log(1+\sin x) = x - x^2/2 + x^3/6 - x^4/12 + \ldots$$

Ex 43 Cauchy's MVT — $f(x)=\sqrt{x}$, $g(x)=1/\sqrt{x}$ on $[1,4]$

Verify Cauchy's MVT for $f(x) = \sqrt{x}$ and $g(x) = 1/\sqrt{x}$ on $[1, 4]$.

STEP 3 – CONDITIONS

Both $\sqrt{x}$ and $1/\sqrt{x}$ are continuous on $[1,4]$ (no zero or negative values). ✓

$f'(x) = 1/(2\sqrt{x})$, $g'(x) = -1/(2x^{3/2})$ — both exist on $(1,4)$. ✓

$g'(x) \neq 0$ for $x \in (1,4)$. ✓

STEP 4 – RHS

$f(1) = 1$, $f(4) = 2 \to f(4)-f(1) = 1$

$g(1) = 1$, $g(4) = 1/2 \to g(4)-g(1) = -1/2$

RHS $= 1 / (-1/2) = -2$

STEP 6 – SET F'(C)/G'(C) = −2

$[1/(2\sqrt{c})] / [-1/(2c^{3/2})] = -2$

$[1/(2\sqrt{c})] \times [-2c^{3/2}] = -2$

$-c^{3/2}/\sqrt{c} = -2$

$-c = -2$

$c = 2$

Check: $1 < 2 < 4$ ✓

✅ **$c = 2 \in (1, 4)$**

Ex 44 Limit — $\lim_{x\to 0} [\tan x - x] / x^3$

Evaluate $\lim_{x\to 0} [\tan x - x] / x^3$ using series expansion.

STEP 4 – EXPAND TAN X

$\tan x = x + x^3/3 + 2x^5/15 + \ldots$

STEP 5 – COMPUTE NUMERATOR

$\tan x - x = (x + x^3/3 + \ldots) - x = x^3/3 + 2x^5/15 + \ldots$

STEP 6 – DIVIDE BY X³ AND TAKE LIMIT

$[\tan x - x] / x^3 = (x^3/3 + 2x^5/15 + \ldots) / x^3 = 1/3 + 2x^2/15 + \ldots$

$\lim_{x\to 0} = 1/3$

$\lim_{x\to 0} [\tan x - x] / x^3 = 1/3$

Ex 45 LMVT Application — $f(x) = x^{1/3}$ on $[-8, 8]$

Can LMVT be applied to $f(x) = x^{1/3}$ on $[-8, 8]$? Find c if applicable.

STEP 3 – CHECK CONDITIONS

Condition 1: $f(x) = x^{1/3}$ is continuous everywhere (cube root of any real number exists). ✓

Condition 2: $f'(x) = (1/3)x^{-2/3} = 1/(3x^{2/3})$

At $x = 0$: $f'(0) = 1/(3\cdot 0) \to$ UNDEFINED (division by zero!)

So f is NOT differentiable at $x = 0 \in (-8, 8)$. ✗

LMVT is NOT applicable!

❌ LMVT is NOT applicable — $f(x) = x^{1/3}$ is not differentiable at $x = 0 \in (-8, 8)$.

📝 Exercise Problems — 50 Questions

Practice questions covering all theorems. Answers provided at end.

🔷 2-Mark Questions — Definitions and Basic Verification

Q1 State Rolle's Theorem and write its three conditions. — 2 Marks

Q2 State Lagrange's Mean Value Theorem. — 2 Marks

Q3 Write the Maclaurin's series for e^x. — 2 Marks

Q4 Write the Maclaurin's series for sin x up to x^5 term. — 2 Marks

Q5 Write the Maclaurin's series for cos x up to x^6 term. — 2 Marks

Q6 Write the Maclaurin's series for log(1+x). — 2 Marks

Q7 State Cauchy's Mean Value Theorem. — 2 Marks

Q8 Is Rolle's Theorem a special case of LMVT? Explain. — 2 Marks

Q9 Define differentiability. Give an example of a function continuous but not differentiable. — 2 Marks

Q10 Write the Lagrange's remainder in Taylor's theorem. — 2 Marks

Q11 What is the geometrical meaning of LMVT? — 2 Marks

Q12 Find the value of c given by Rolle's theorem for $f(x) = x^2 - 4$ on $[-2, 2]$. — 2 Marks

Q13 Find the value of c given by LMVT for $f(x) = x^2$ on $[0, 2]$. — 2 Marks

Q14 Evaluate: lim(x→0) [(e^x − 1)/x] using series. — 2 Marks

Q15 Write the expansion of (1+x)^n for small x. — 2 Marks

🔶 5-Mark Questions — Full Theorem Applications

Q16 Verify Rolle's theorem for $f(x) = x^2 - 3x + 2$ on $[1, 2]$. — 5 Marks

Q17 Verify Rolle's theorem for $f(x) = x^3 - x$ on $[-1, 1]$. — 5 Marks

Q18 Verify Rolle's theorem for $f(x) = \sin^2 x$ on $[0, \pi]$. — 5 Marks

Q19 Verify Rolle's theorem for $f(x) = x(x+3)e^{(-x/2)}$ on $[-3, 0]$. 5 Marks

Q20 Verify Rolle's theorem for $f(x) = (x^2-1)(x-2)$ on $[1, 2]$. 5 Marks

Q21 Verify LMVT for $f(x) = x^3$ on $[0, 1]$. 5 Marks

Q22 Verify LMVT for $f(x) = e^x$ on $[0, 1]$. 5 Marks

Q23 Verify LMVT for $f(x) = x^2 + 2x + 3$ on $[4, 6]$. 5 Marks

Q24 Verify LMVT for $f(x) = \log x$ on $[1, 3]$. 5 Marks

Q25 Using LMVT, prove that $(b-a)/b < \log(b/a) < (b-a)/a$ for $b > a > 0$. 5 Marks

Q26 Verify Cauchy's MVT for $f(x) = x^2$, $g(x) = x$ on $[1, 2]$. 5 Marks

Q27 Verify Cauchy's MVT for $f(x) = \sin x$, $g(x) = x$ on $[0, \pi/2]$. 5 Marks

Q28 Verify Cauchy's MVT for $f(x) = x^3$, $g(x) = x^2$ on $[1, 3]$. 5 Marks

Q29 Expand $e^{(2x)}$ using Maclaurin's series up to x^4 term. 5 Marks

Q30 Expand $\sin(2x)$ using Maclaurin's series up to x^5 term. 5 Marks

Q31 Expand $\log(1-x)$ using Maclaurin's series. 5 Marks

Q32 Expand $e^x \sin x$ using Maclaurin's series up to x^4 term. 5 Marks

Q33 Expand $(1+x)^{(1/2)}$ using binomial series up to x^3 term. 5 Marks

Q34 Expand $\cos^2 x$ using Maclaurin's series up to x^6 term. 5 Marks

Q35 Expand $e^x / (1-x)$ using Maclaurin's series up to x^3 term. 5 Marks

Q36 Expand $f(x) = \sqrt{(1+x)}$ in powers of $(x-3)$ using Taylor's series up to 3rd degree. 5 Marks

Q37 Expand e^x in powers of $(x-1)$ up to 4th degree term. 5 Marks

Q38 Expand $\sin x$ in powers of $(x - \pi/2)$ up to 3rd degree term. 5 Marks

Q39 Find approximate value of $e^{0.2}$ using Maclaurin's series (4 decimal places). 5 Marks

Q40 Find approximate value of $\cos 46°$ using Taylor's expansion about $45°$. 5 Marks

Q41 Find approximate value of $\log(1.2)$ using Maclaurin's series. 5 Marks

Q42 Find approximate value of $\sqrt{(1.04)}$ using binomial expansion. 5 Marks

Q43 Using LMVT, prove that $n \cdot b^{(n-1)}(a-b) \le a^n - b^n \le n \cdot a^{(n-1)}(a-b)$ for $a > b > 0$, $n > 1$. 5 Marks

Q44 Estimate the error in computing sin(0.1) using the first two non-zero terms of its series. 5 Marks

Q45 Estimate the error in computing log(1.1) using 3 terms of its Maclaurin series. 5 Marks

Q46 Expand $\log(1 + x + x^2)$ using Maclaurin's series up to x^4 term. 5 Marks

Q47 Evaluate $\lim(x\to 0)\ [x - \sin x] / x^3$ using series. 5 Marks

Q48 Evaluate $\lim(x\to 0)\ [\log(1+x) - x] / x^2$ using series. 5 Marks

Q49 Expand $e^{(-x^2)}$ using Maclaurin's series up to x^8 term. 5 Marks

Q50 Can Rolle's theorem be applied to f(x) = tan x on [0, π]? Give full justification. 5 Marks

💡 Selected Answers / Hints

Q#	Answer / Hint
Q12	$c = 0$
Q13	$c = 1$
Q14	1
Q16	$c = 3/2$
Q17	$c = \pm 1/\sqrt{3}$
Q21	$c = 1/\sqrt[3]{3}$
Q22	$c = \log(e-1) \approx 0.541$
Q24	$c = 2/\log 3$
Q39	≈ 1.2214
Q41	≈ 0.1823
Q42	≈ 1.0198
Q47	$1/6$
Q48	$-1/2$

Q50	Not applicable — $\tan x$ is discontinuous at $x = \pi/2 \in (0,\pi)$

📋 Complete Formula Sheet & Quick Revision Notes

▶ THEOREM FORMULAS

Rolle's Theorem

Conditions: f continuous on $[a,b]$, differentiable on (a,b), $f(a)=f(b)$

Conclusion: $\exists\, c \in (a,b) : f'(c) = 0$

Lagrange's MVT

Conditions: f continuous on $[a,b]$, differentiable on (a,b)

$\exists\, c \in (a,b) : f'(c) = [f(b)-f(a)]/(b-a)$

Cauchy's MVT

Conditions: f,g continuous on $[a,b]$, differentiable on (a,b), $g'(x)\neq 0$

$\exists\, c \in (a,b) : f'(c)/g'(c) = [f(b)-f(a)]/[g(b)-g(a)]$

Taylor's Series

$f(x) = f(a) + (x-a)f'(a) + (x-a)^2/2!\ f''(a) + (x-a)^3/3!\ f'''(a) + \ldots$

Remainder: $R_n = (x-a)^{(n+1)}/(n+1)! \times f^{(n+1)}(\xi)$

Maclaurin's Series

$f(x) = f(0) + xf'(0) + x^2/2!\ f''(0) + x^3/3!\ f'''(0) + \ldots$

(Taylor's at $a=0$)

Remainder/Error Bound

$|R_n| \leq M\cdot|x-a|^{(n+1)}/(n+1)!$

$M = \max|f^{(n+1)}(x)|$ on $[a,b]$

▶ STANDARD EXPANSIONS

Function	Maclaurin Expansion	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$	All x
$e^{(-x)}$	$1 - x + x^2/2! - x^3/3! + x^4/4! - \ldots$	All x
$\sin x$	$x - x^3/3! + x^5/5! - x^7/7! + \ldots$	All x
$\cos x$	$1 - x^2/2! + x^4/4! - x^6/6! + \ldots$	All x
$\tan x$	$x + x^3/3 + 2x^5/15 + \ldots$	$\|x\| < \pi/2$
$\log(1+x)$	$x - x^2/2 + x^3/3 - x^4/4 + \ldots$	$-1 < x \leq 1$
$\log(1-x)$	$-x - x^2/2 - x^3/3 - x^4/4 - \ldots$	$-1 \leq x < 1$

(1+x)^n	$1 + nx + n(n-1)x^2/2! + \dots$	$|x| < 1$
1/(1+x)	$1 - x + x^2 - x^3 + \dots$	$|x| < 1$
sinh x	$x + x^3/3! + x^5/5! + \dots$	All x
cosh x	$1 + x^2/2! + x^4/4! + \dots$	All x

▶ USEFUL DERIVATIVES TO REMEMBER

Function	Derivative	Notes
x^n	n·x^(n−1)	Power rule
e^x	e^x	Same function!
e^(ax)	a·e^(ax)	Chain rule
log x	1/x	Natural log
sin x	cos x	
cos x	−sin x	Note the minus sign!
tan x	$\sec^2 x$	
u·v	u'v + uv'	Product rule
u/v	$(u'v - uv')/v^2$	Quotient rule

Quick Revision Notes

Last-Minute Review Before Your Exam

✅ Key Conditions Checklist

- ✓ Rolle's: Continuous [a,b] + Differentiable (a,b) + f(a)=f(b)
- ✓ LMVT: Continuous [a,b] + Differentiable (a,b)
- ✓ Cauchy's: Above + $g'(x) \neq 0$ in (a,b)
- ✓ Taylor/Maclaurin: Function must be infinitely differentiable

❌ Common Mistakes to Avoid

- ✗ Checking continuity on (a,b) instead of [a,b]
- ✗ Applying Rolle's without checking f(a)=f(b)
- ✗ Forgetting that c must lie in OPEN interval (a,b)
- ✗ Using LMVT formula wrong: slope of chord, not of tangent at a or b
- ✗ Expanding e^x with wrong signs (all signs are positive!)
- ✗ Confusing sin x (odd powers) with cos x (even powers)

💡 Memory Tips

- **Rolle's = Rollercoaster** that starts and ends at same height — must have a flat top/bottom!
- **LMVT = Average speed** — there's always an instant where speed = average
- **sin x series**: x, x^3, x^5 (odd) | alternating: +, −, +, −
- **cos x series**: $1, x^2, x^4$ (even) | alternating: +, −, +, −
- **log(1+x)**: No factorials! Denominators are natural numbers
- **e^x**: All positive, all factorials

📌 Key Relationships

- Rolle's theorem ⊂ LMVT (special case when f(a)=f(b))
- LMVT ⊂ Cauchy's (special case when g(x)=x)
- Maclaurin ⊂ Taylor (special case when a=0)
- If ALL derivatives of f at 0 are known → use Maclaurin
- If expanding near $a \neq 0$ → use Taylor

🔢 Approach for Exam Questions

Problem Type	Approach
"Verify Rolle's / LMVT / Cauchy's"	Check conditions → Write formula → Differentiate → Set equal → Solve → Verify c ∈ (a,b)
"Expand f(x) using Maclaurin's"	Find f(0), f'(0), f''(0)... → Substitute into formula
"Expand f(x) about x = a"	Find f(a), f'(a), f''(a)... → Use Taylor's formula
"Find approximate value"	Use appropriate expansion → Substitute x value → Add terms

"Evaluate limit"	Expand numerator/denominator → Cancel x terms → Take limit
"Prove inequality"	Apply LMVT to suitable function → Use sign of derivative

🎯 You Are Ready!

You have studied all theorems, worked through 45+ solved examples, and reviewed formulas.

Remember: **Understand the conditions, verify step by step, and never skip a check.**

Best of luck in your examination!